

DALHART, TX, USA

WMO: 747360

Lat: 36.017N

Lon: 102.550W

Elev: 1216

StdP: 87.54

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93042

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-12.7	-10.1	-18.7	0.8	-3.4	-16.8	1.0	-1.2	16.4	2.6	14.6	2.9	4.5	10	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	37.3	18.2	35.9	18.3	34.4	18.3	21.2	30.4	20.6	30.0	20.1	29.6	6.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	15.9	23.2	18.0	15.1	22.8	17.6	14.6	22.6	67.3	30.7	65.0	29.6	62.9	29.4	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	12.0	10.7	DB	-17.0	39.8	2.6	1.3	-18.9	40.8	-20.4	41.6	-21.9	42.3	-23.7	43.3	(4)
(5)			WB	-17.5	22.8	2.4	1.9	-19.3	24.1	-20.7	25.2	-22.0	26.2	-23.7	27.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	2.4	4.0	8.3	12.6	18.0	23.7	25.7	24.7	20.8	13.9	7.4	2.3		
	DBStd	9.56	5.06	5.64	5.32	4.74	4.37	3.29	2.42	2.57	3.84	4.76	5.00	5.15		
	HDD10.0	908	238	177	94	28	3	0	0	0	1	19	106	242		
	HDD18.3	2443	493	402	313	177	59	2	1	1	21	150	328	498		
	CDD10.0	2258	3	9	41	106	251	412	487	457	324	139	28	3		
	CDD18.3	752	0	0	1	5	49	164	229	198	94	11	0	0		
	CDH23.3	9921	1	11	75	254	864	2118	2843	2262	1176	287	30	1		
	CDH26.7	5049	0	2	14	69	382	1146	1602	1198	540	93	2	0		
(14) Wind	WSAvg	5.3	4.9	5.3	5.7	6.2	5.7	5.7	5.0	4.6	5.0	5.1	5.1	5.0		
(15) Precipitation	PrecAvg	456	11	11	24	36	63	61	72	73	37	27	18	12		
	PrecMax	678	35	45	94	115	144	178	191	248	87	144	81	57		
	PrecMin	279	0	0	0	0	4	0	10	2	0	0	0	0		
	PrecStd	108	11	13	25	31	39	41	42	50	26	35	21	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.4	24.8	28.6	31.3	35.9	39.3	39.4	37.8	36.2	31.8	26.5	21.2		
		MCWB	8.6	9.7	11.2	12.4	14.8	16.6	18.7	18.6	17.4	14.6	11.3	8.5		
	2%	DB	18.4	21.3	25.5	28.6	33.2	37.3	37.3	36.1	34.0	29.2	23.8	17.7		
		MCWB	7.0	8.3	10.3	11.8	14.5	17.0	19.0	19.0	17.2	14.2	10.4	7.1		
	5%	DB	15.3	18.4	22.6	26.5	31.1	35.2	35.8	34.6	32.2	27.0	20.8	15.0		
		MCWB	5.9	7.2	9.5	11.6	14.6	17.6	19.0	19.2	17.3	13.6	9.4	6.0		
	10%	DB	12.2	14.9	19.5	23.8	28.6	33.1	34.1	32.9	30.0	23.8	17.4	11.9		
		MCWB	4.5	5.7	8.5	11.0	14.5	17.8	19.1	19.2	17.1	12.7	8.1	4.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	10.5	13.7	15.7	19.1	21.5	21.9	22.1	20.6	18.7	13.0	9.7		
		MCDB	19.5	21.3	21.8	23.1	27.8	31.4	31.3	30.4	29.4	24.3	20.5	17.4		
	2%	WB	7.7	9.0	11.6	14.3	17.9	20.4	21.2	21.3	19.7	16.9	11.5	7.9		
		MCDB	17.0	19.6	21.4	22.4	26.7	30.4	30.5	30.1	28.1	22.3	20.3	17.0		
	5%	WB	6.3	7.7	10.4	13.0	16.9	19.8	20.6	20.6	18.9	15.4	10.1	6.3		
		MCDB	14.8	17.6	20.7	22.0	25.5	29.6	30.0	29.7	27.4	22.2	18.8	14.1		
	10%	WB	4.7	6.2	9.1	11.8	15.9	19.1	20.1	20.1	18.2	14.0	8.8	4.9		
		MCDB	11.8	14.4	18.8	21.7	24.2	28.5	30.0	29.2	26.4	21.8	17.0	11.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	16.2	17.1	16.8	16.2	16.1	15.5	14.8	15.4	16.1	16.3	15.0		
		MCDBR	21.0	22.4	22.3	21.9	20.4	19.7	18.3	17.6	19.0	21.0	21.7	20.1		
		MCWBR	11.8	11.8	10.3	9.1	7.5	6.1	5.0	5.0	6.2	8.7	10.8	11.3		
	5% WB	MCDBR	20.0	20.5	19.9	18.1	16.6	16.1	15.1	14.7	15.1	15.5	18.9	19.0		
		MCWBR	11.5	11.1	9.9	8.7	7.2	5.8	4.8	4.8	5.4	7.5	9.8	11.1		
(40) Clear-Sky Solar Irradiance	taub		0.250	0.255	0.287	0.312	0.331	0.362	0.376	0.362	0.328	0.293	0.265	0.250		
	taud		2.583	2.589	2.487	2.430	2.434	2.389	2.400	2.416	2.488	2.558	2.557	2.582		
	Ebn at Noon		974	1004	987	971	950	915	901	909	929	943	944	952		
	Edh at Noon		74	83	101	113	115	120	118	113	100	85	75	69		
(44) All-Sky Solar Radiation	RadAvg		3.07	3.93	5.24	6.46	7.30	7.75	7.48	6.64	5.68	4.33	3.31	2.66		
	RadStd		0.24	0.32	0.39	0.35	0.43	0.37	0.31	0.32	0.28	0.40	0.28	0.15		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+89	N/A

Nomenclature: See separate page